Rishabh Mehra

Software Engineer (New Grad) | Backend & Full Stack
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Summary

Recent Computer Science graduate from the University of Alberta with hands-on experience across backend systems, cloud infrastructure, and full-stack development, built through co-ops in research computing, IT operations, and software engineering. Strong foundation in computer science fundamentals, with hands on experience working in Git-based, collaborative team environments. Open to feedback and quick to pick up new tools and codebases.

Education

University of Alberta April 2026
B.Sc. Specialization in Computer Science *Edmonton, AB*
- Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Databases, Software Engineering, Object-Oriented Design, App Development

Experience

University of Alberta Edmonton, AB
Junior Research Analyst Co-op *May 2025 – Aug 2025*
- Automated diagnostics and monitoring scripts across 50+ Linux systems using Python and Bash, reducing manual troubleshooting time and improving environment stability.
- Provisioned and maintained OpenStack-based virtual machines to support reliable execution of compute-intensive research workflows, documenting procedures for team-wide reuse.

Elk Valley Resources Elkford, BC
Desktop Support Engineer Co-op *Sep 2024 – Apr 2025*
- Developed a centralized inventory platform for 15+ IT staff across 5 mine sites, applying OOP design principles and tiered access controls to replace a fragmented manual process and improve visibility over 50+ hardware assets.
- Resolved incident tickets and automated ServiceNow work order generation for inter-site stock transfers using custom scripts and business rules, eliminating 30+ manual tickets and saving 10+ hours per week in operational overhead.

Marquee Equity New Delhi, India
Software Engineer Intern *May 2023 – July 2023*
- Analyzed 1M+ records of investor and startup data using Python and SQL, cleaning and joining datasets to surface patterns that improved investor-startup matching accuracy.
- Prepared executive summaries and findings reports, presenting analysis in a clear and structured way to support decision making across teams.

Projects

DocChat | *Python, Vector Databases, RAG, LangChain, LangGraph, Multi Agent Systems* July 2026 – Aug 2026
- Architected a multi-agent RAG pipeline that pairs BM25 keyword search with vector embeddings for hybrid retrieval, capturing exact terminology and semantic meaning across long, table-heavy documents.
- Built a verification and self-correction loop that cross-checks LLM-generated answers against source documents, catching unsupported claims and flagging out-of-scope questions before they reach the user.

Core Banking Ledger Service | *Java, Spring Boot, Spring Security, Redis, Kafka, Docker* July 2026 – Aug 2026
- Engineered a double-entry accounting ledger core enforcing balanced debit/credit transactions, preventing duplicate transactions between accounts using idempotent RESTful transfer APIs.
- Architected a decoupled monitoring service that independently analyzes transaction activity as it occurs, isolating fraud detection logic from core transfer processing so that a slowdown or failure in one system never affects the other.

Forget Me Not (Care Companion) | *React Native, TypeScript, Django, PostgreSQL, AWS* Sep 2025 – Nov 2025
- Built a cross-platform single-page companion-matching application for elderly users, designing a PostgreSQL data model handling user profiles, caregiver verifications, real-time chat, and Stripe payment records.
- Delivered GPS safety tracking and an admin compliance dashboard, integrating third-party APIs and deploying on AWS for a secure, scalable production environment.

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL
Frontend: ReactJS, React Native, HTML, CSS
Backend & Frameworks: Django, Node.js, Express, SpringBoot
Cloud & Databases: AWS (EC2, RDS), Github Actions, GCP, Docker, PostgreSQL, Firebase
Tools & Practices: Git, CI/CD, Linux, Agile/Scrum, OOP, LLM Integration (Gemini API)